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# Understanding the Blinkit delivery data

A comprehensive guide to the five sheets in Blinkit_Dataset.xlsx, how they connect, what the dashboard shows, and what the data can and cannot support.

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## Summary

1,500 orders from 20 stores in 5 cities, 1–14 September 2026. The North Star (the on-time fulfilment rate) is 73.7%. It is provisional, meaning it may change: 19 cancellations have no confirmed cause, and counting them would bring it to 72.8%. The clearest problem is severe delays at peak: from 7 to 10 PM, 25.0% of deliveries are more than 5 minutes late vs 11.8% off-peak, while promises get shorter at peak. These are patterns in these orders, not proof of cause, and they don't describe the whole Blinkit network.


## How to read the evidence

- **Recorded**: timestamps, status, cancellation reason, distance, rain (from the data).
- **Estimated**: picking, rider-wait and travel minutes (supplied as estimates).
- **Correlation**: two measures moving together in a straight line, shown with a 95% CI (the range of values that fit the data). Section 8 explains each term with an example.
- **Unproven cause**: every explanation in section 7 is a hypothesis.

## Contents


1. The five sheets
2. Data dictionary
3. How the sheets connect
4. Checks and decisions
5. Definitions, formulas and the issue tree
6. How to read the dashboard
7. Findings, hypotheses and limitations
8. Key concepts and methodology
9. Key questions and answers

# 1. The five sheets

What each sheet contains, what one row means, and its limits.

## Orders

**1,500 rows · 25 columns**

| | |
|---|---|
| One row = | One order, delivered or cancelled. |
| ID column | `order_id` e.g. `O00001` |
| Main contents | Placed, promised and delivered times; status and cancellation reason; estimated picking, rider-wait and travel minutes; distance; rain; event day; traffic. |
| Limits | Stage durations are estimates. Rider and delivery fields are blank on the 45 cancelled orders. Holds 1.2% of the orders counted in Demand. |

## Stores

**20 rows · 12 columns**

| | |
|---|---|
| One row = | One store. |
| ID column | `store_id` e.g. `S001` |
| Main contents | City, age, staffing per shift, item availability, average picking time, typical orders per hour, radius, and summary delivery figures. |
| Limits | One figure per store, not per hour. |

## Riders

**80 rows · 9 columns**

| | |
|---|---|
| One row = | One rider. |
| ID column | `rider_id` e.g. `R001` |
| Main contents | Home store, vehicle, experience, typical trips, distance, idle time, acceptance rate, store wait. |
| Limits | Home store is not where the rider worked in a given hour. No hourly log-ins. |

## Demand

**4,480 rows · 8 columns**

| | |
|---|---|
| One row = | Orders placed in one store in one hour. |
| ID column | `store_id + date + hour` e.g. `S001 · 2026-09-01 · 8` |
| Main contents | Orders placed per store-hour, 8 AM to 11 PM, with peak window, rain and event day. |
| Limits | Counts only; no outcomes. |

## Customers

**250 rows · 10 columns**

| | |
|---|---|
| One row = | One customer. |
| ID column | `customer_id` e.g. `C0001` |
| Main contents | City, sign-up date, delivered orders, average rating, tickets and main reason, refunds, repeat flags. |
| Limits | Tickets and refunds are per customer, not per order. |

# 2. Data dictionary

All 64 columns, exactly as named in the workbook.

## Orders

| Column | Plain meaning |
|---|---|
| `order_id` | Unique order ID |
| `customer_id` | Customer who placed the order |
| `store_id` | Store that fulfilled the order |
| `rider_id` | Rider who delivered it (blank if cancelled) |
| `placed_at` | When the order was placed |
| `promised_delivery_at` | Delivery time promised at checkout (the deadline) |
| `actual_delivery_at` | When it was delivered (blank if cancelled) |
| `order_status` | Delivered or Cancelled |
| `basket_size` | Basket size category |
| `item_count` | Number of items |
| `order_value_inr` | Order value in rupees |
| `distance_km` | Store-to-customer distance |
| `picking_time_min_est` | Estimated picking time (estimate) |
| `rider_wait_time_min_est` | Estimated rider wait at store (estimate) |
| `travel_time_min_est` | Estimated travel time (estimate) |
| `promised_eta_min` | Promised minutes = promised − placed |
| `actual_delivery_duration_min` | Actual minutes = delivered − placed |
| `delay_min` | Actual − promised minutes (positive = late) |
| `on_time_flag` | 1 if delay ≤ 0 |
| `cancellation_reason` | Why a cancelled order was cancelled |
| `order_hour` | Hour the order was placed |
| `is_peak_hour` | True for orders placed 7–10 PM |
| `rain_condition` | Light or Heavy; blank = no rain |
| `event_day` | True on event days |
| `traffic_congestion_index` | Traffic level, 0 to 1 |

## Stores

| Column | Plain meaning |
|---|---|
| `store_id` | Unique store ID |
| `city` | City |

| Column | Plain meaning |
|---|---|
| `store_age_days` | Days since opening |
| `staffing_per_shift` | Staff per shift |
| `item_availability_pct` | Share of items in stock |
| `avg_picking_time_min` | Average picking time |
| `typical_orders_per_hour` | Usual orders per hour |
| `service_radius_km` | Delivery radius |
| `delivered_orders` | Delivered orders (summary) |
| `avg_actual_delivery_duration_min` | Average delivery minutes (summary) |
| `late_orders` | Late orders (summary) |
| `on_time_rate_pct` | On-time rate (summary) |

## Riders

| Column | Plain meaning |
|---|---|
| `rider_id` | Unique rider ID |
| `home_store_id` | Rider's home store (not hourly location) |
| `vehicle_type` | EV scooter, bicycle or petrol scooter |
| `experience_months` | Months of experience |
| `typical_trips_per_shift` | Usual trips per shift |
| `avg_distance_per_trip_km` | Average trip distance |
| `avg_idle_time_min_per_shift` | Average idle minutes per shift |
| `acceptance_rate_pct` | Share of offered orders accepted |
| `avg_store_wait_time_min` | Average wait at store (rider level) |

## Demand

| Column | Plain meaning |
|---|---|
| `store_id` | Store |
| `date` | Date |
| `hour` | Hour (8–23) |
| `day_of_week` | Weekday name |
| `orders_placed_in_hour` | All orders placed in that store-hour |
| `is_peak_window` | True for 7–10 PM |
| `rain_condition` | Light or Heavy; blank = no rain |
| `event_day` | True on event days |

## Customers

| Column | Plain meaning |
|---|---|
| `customer_id` | Unique customer ID |
| `city` | City |
| `signup_date` | Sign-up date |
| `orders_in_dataset` | Delivered orders in the Orders sheet |
| `avg_delivery_rating_1_to_5` | Average delivery rating |
| `repeat_customer_flag` | Repeat customer overall |
| `support_tickets_count` | Number of support tickets |
| `primary_ticket_reason` | Main ticket reason (blank if none) |
| `refunds_and_goodwill_credits_inr` | Refunds and credits in rupees |
| `repeat_order_in_dataset_flag` | Ordered more than once in the period |

# 3. How the sheets connect

Each row below shows how a row in the Orders sheet finds matching information in another sheet. Every key was matched row by row. These links connect information between sheets; they are not the steps of a delivery. (The delivery stages, picking, rider wait and travel, are estimated columns inside the Orders sheet.)

**Orders**
1,500 rows
`order_id`

**Many orders can come from one store.**
Example: store S010 (Delhi) supplied 72 orders, including O00001 and O00069.
Matched on `store_id`

**Stores**
20 rows

**Orders**
1,500 rows
`order_id`

**One rider can deliver several orders.**
Example: rider R003 delivered 14 orders, including O00001 and O00005.
Matched on `rider_id`

**Riders**
80 rows

**Orders**
1,500 rows
`order_id`

**One customer can place several orders.**
Example: customer C0113 placed 5 orders, including O00001 and O00394.
Matched on `customer_id`

**Customers**
250 rows

**Orders**
1,500 rows
`order_id`

**Orders from the same store, date and hour share one Demand record.**
Example: O00002 and O00843 were both placed at store S008 on 9 Sep, 18:00–18:59. They share one Demand row, which counts 18 orders placed in that hour.
Matched on `store_id + date + hour`

**Demand**
4,480 store-hours

All 1,500 orders stay in, including the 45 cancelled orders that have no rider recorded. Their rider details are left blank.

## Matching audit

| Link | Key in first sheet | Key in second sheet | Rows with a key | Matched | Not matched | Note |
|---|---|---|---|---|---|---|
| **Orders → Stores** | `store_id` | `store_id` | 1,500 | 1,500 | 0 | Many orders to one store. |
| **Orders → Riders** | `rider_id` | `rider_id` | 1,455 | 1,455 | 0 | Rider ID is blank on the 45 cancelled orders. No rider ID is recorded. This does not prove that a rider was never assigned. |
| **Orders → Customers** | `customer_id` | `customer_id` | 1,500 | 1,500 | 0 | Many orders to one customer. |
| **Orders → Demand** | `store_id + date + order_hour` | `store_id + date + hour` | 1,500 | 1,500 | 0 | Each order finds its store-hour. Demand counts all orders placed in that hour; the Orders sheet holds only some of them. |
| **Riders → Stores (home store)** | `home_store_id` | `store_id` | 80 | 80 | 0 | Home store is a rider attribute only; riders deliver for many stores. |

**Join type.** Left joins from Orders keep all 1,500 orders. Cancelled orders keep a blank rider rather than being dropped.

## Orders vs Demand: not the same population

- Demand counts 128,528 orders placed in 4,480 store-hours. The Orders sheet holds 1,500 of them (1.2%).
- Orders fall in 1,264 of the 4,480 store-hours, and busier hours don't get more sampled orders (r = 0.02).
- Peak hours are 27.9% of Demand but 18.8% of Orders.
- So the Orders sheet does not mirror Demand hour by hour. Whether these results apply more widely depends on how the sample was selected. That information is missing. Until it is known, results describe these orders only.

# 4. Checks and decisions

The original workbook was never changed. Nothing missing was replaced with zero.

| Item | Scope | What was done | Why |
|---|---|---|---|
| Blank rain_condition | 1,063 orders, 3,214 Demand rows | Treated as no rain | Confirmed by the mentor. |
| Timing columns | All 1,455 delivered orders | Checked delay = actual − promised, promise = promised − placed, on_time_flag = delay ≤ 0 | All agree. |
| Blank rider and delivery fields | 45 cancelled orders | Kept blank, not zero | No rider ID is recorded. This does not prove that a rider was never assigned. These orders were not delivered. |
| Unclear cancellations | 19 orders | Kept outside E; North Star shown with a range | The recorded category does not establish the underlying cause. |
| Delivery completion | All delivered orders | Treated actual_delivery_at as confirmed customer handover and promised_delivery_at as the original checkout deadline | Assumption to confirm: the data does not say how these times are recorded. |
| Customer order counts | 42 customers | No change | orders_in_dataset counts delivered orders; the gap equals the cancellations. |
| Rider home store | 5.3% of orders match | Not used as hourly availability | Riders work across stores. |
| Stage durations | All delivered orders | Labelled as estimates | Not recorded timestamps. |
| Store summaries | 20 stores | Recalculated from Orders | Match exactly. |
| Orders vs Demand | 1.2% coverage | Results limited to these orders | Whether these results apply more widely depends on how the sample was selected. That information is missing. Peak hours are also under-represented. |
| Duplicates and IDs | All sheets | Checked | No duplicate IDs; every key matches. |

# 5. Definitions and formulas

The rules behind every number. Results are for all 1,500 orders.

- **On time**: handed over to the customer at or before the original checkout deadline. Exactly on the deadline counts as on time. *Assumption to confirm:* `actual_delivery_at` records the confirmed handover and `promised_delivery_at` is the original promise, not an updated ETA.
- **Peak**: order placed 7–10 PM (`is_peak_hour`); all other hours are off-peak.
- **Company-caused cancellation**: the recorded reason points to a failure on Blinkit's side: rider unavailable (18) or item unavailable (8). The underlying reason decides this, not who pressed Cancel. **Unresolved**: payment issue (8) and customer cancelled (11). "Customer cancelled" is only the recorded category; it does not establish the cause, or even for certain who pressed Cancel.
- **Percentages in the issue tree**: each group's order count ÷ 1,481 × 100. Fields used from the Orders sheet: order_status, delay_min, is_peak_hour and cancellation_reason. Counts add up exactly; child percentages may differ from the parent by 0.1 because of rounding.

## Formulas

"Runnable" formulas work in Excel 365 once the Orders sheet is formatted as an Excel table named `Orders`. "Illustrative" formulas show the calculation; replace the names with your own cells or ranges before running them.

| Measure | In words | Formula | Type | Result |
|---|---|---|---|---|
| **Minutes late (one order)** | Delivered time minus promised time, in minutes. Excel stores times as days, so the difference is multiplied by 1,440 (minutes in a day). | `=(Orders[@actual_delivery_at]-Orders[@promised_delivery_at])*1440` | Runnable | Matches delay_min for every delivered order |
| **Late orders** | Delivered orders with minutes late above 0 | `=COUNTIFS(Orders[order_status],"Delivered",Orders[delay_min],">0")` | Runnable | 363 |
| **Severely late orders** | Delivered orders more than 5 minutes late (our analytical cut-off) | `=COUNTIFS(Orders[order_status],"Delivered",Orders[delay_min],">5")` | Runnable | 208 |
| **On-time orders** | Delivered orders with minutes late of 0 or less | `=COUNTIFS(Orders[order_status],"Delivered",Orders[delay_min],"<=0")` | Runnable | 1,092 |
| **Eligible orders (E)** | Delivered orders + rider-unavailable cancellations + item-unavailable cancellations, as one total | `=COUNTIFS(Orders[order_status],"Delivered")+COUNTIFS(Orders[cancellation_reason],"Rider unavailable")+COUNTIFS(Orders[cancellation_reason],"Item unavailable")` | Runnable | 1,455 + 18 + 8 = 1,481 |
| **North Star (provisional)** | On-time orders ÷ E × 100 | `=on_time/eligible*100 (replace the two names with the cells holding the counts above)` | Illustrative | 1,092 ÷ 1,481 × 100 = 73.7% |
| **North Star, lower end** | As if all unclear cancellations were company-caused: on-time orders ÷ all orders × 100 | `=on_time/ROWS(Orders[order_id])*100 (replace on_time with its cell)` | Illustrative | 1,092 ÷ 1,500 × 100 = 72.8% |

| Measure | In words | Formula | Type | Result |
|---|---|---|---|---|
| **Cancellation rate** | Cancelled orders ÷ all orders placed × 100 | `=COUNTIFS(Orders[order_status],"Cancelled")/ROWS(Orders[order_id])*100` | Runnable | 45 ÷ 1,500 × 100 = 3.0% |
| **Average delivery time** | Mean actual minutes of delivered orders (blank for cancelled orders) | `=AVERAGEIFS(Orders[actual_delivery_duration_min],Orders[order_status],"Delivered")` | Runnable | 11.31 min |
| **P90 delivery time** | 90% of delivered orders are at or below this (blank cells are ignored) | `=PERCENTILE.INC(Orders[actual_delivery_duration_min],0.9)` | Runnable | 17.2 min |
| **Average promise** | Mean promised minutes of delivered orders | `=AVERAGEIFS(Orders[promised_eta_min],Orders[order_status],"Delivered")` | Runnable | 10.33 min |
| **Load ratio (one order)** | Orders placed in that store-hour (Demand) ÷ the store's typical orders per hour (Stores) | `orders_placed_in_hour ÷ typical_orders_per_hour, after looking both up for the order's store, date and hour` | Illustrative | Every peak hour ≥ 1.05 |
| **Difference and 95% CI** | Gap between two percentages, with the range of differences that fit the data | `p1 − p2 ± 1.96 × √(p1(1−p1)/n1 + p2(1−p2)/n2), with p as proportions` | Illustrative (maths, not an Excel formula) | Severe, peak − off-peak: +13.2 pts (95% CI +7.8 to +18.6) |
| **Correlation (r)** | How closely two columns rise or fall together in a straight line (−1 to +1) | `=CORREL(range1,range2) (replace with the two column ranges, e.g. the 20 stores' picking times and late rates)` | Illustrative | Store picking time vs late rate: r = 0.43 |

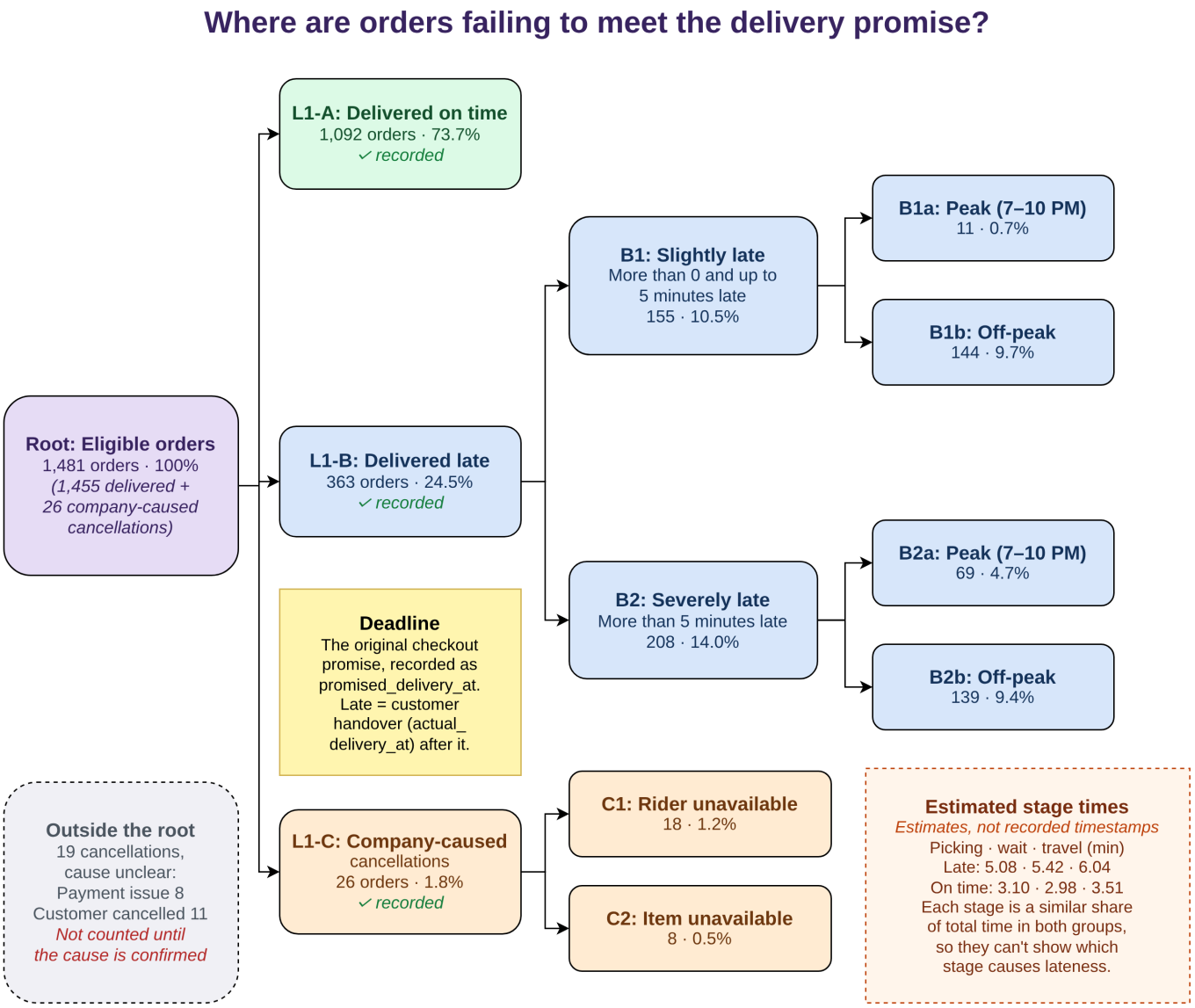
## Issue tree counts

| Box | Orders | % of E |
|---|---|---|
| Root: eligible orders | 1,481 | 100.0% |
| L1-A Delivered on time | 1,092 | 73.7% |
| L1-B Delivered late | 363 | 24.5% |
| B1 Slightly late (0–5 min) | 155 | 10.5% |
| B1a Peak | 11 | 0.7% |
| B1b Off-peak | 144 | 9.7% |
| B2 Severely late (>5 min) | 208 | 14.0% |
| B2a Peak | 69 | 4.7% |
| B2b Off-peak | 139 | 9.4% |
| L1-C Company-caused cancellations | 26 | 1.8% |
| C1 Rider unavailable | 18 | 1.2% |
| C2 Item unavailable | 8 | 0.5% |

Outside the root (not part of the 1,481): 19 cancellations with unconfirmed cause (payment issue 8, customer cancelled 11).

# 5b. The issue tree

The same tree as Slide 2 of the submission, generated from the draw.io file.




- **North Star (provisional)** = L1-A ÷ (L1-A + L1-B + L1-C) × 100 = 1,092 ÷ 1,481 = 73.7%. If all 19 unclear cancellations were company-caused: 72.8%. On time among delivered orders: 75.1%.
- **Each box's %** = its order count ÷ 1,481 × 100. Counts add up exactly; a child % may differ from its parent by 0.1 because of rounding.
- **Orders sheet fields used:** order_status, delay_min, is_peak_hour (peak = placed 7–10 PM) and cancellation_reason.
- **Severe** (more than 5 minutes late) is our analytical choice, not a dataset label; the peak pattern also holds at 3 and 7 minutes.
- **Scope:** 1,500 orders, 20 stores, 1–14 Sep 2026. Selection method not recorded, so results describe these orders only. Possible causes: Slide 3.

# 6. How to read the dashboard

Eleven tabs. Every KPI has a yellow "source" chip: tap or hover on it to see its sheet, column and calculation. Underlined words open a short explanation the same way, and every chart states what it shows, what its percentages are out of, and which sheet it uses.

| Tab | What it shows |
|---|---|
| **How the five sheets connect** | Each sheet, the relationship map, the matching audit, an order lookup across all five sheets, and where each number comes from. |
| **Delivery performance** | KPIs, the issue-tree groups, late and severe rates by hour, daily trend, cancellations, rain, distance, city and vehicle. Filters recalculate everything. |
| **Estimated stage times** | Average picking, rider-wait and travel estimates for late, on-time, peak and off-peak orders. |
| **Stores** | The 20 stores with late rates recalculated from Orders. |
| **Riders and demand** | Orders vs Demand by hour, rider experience, and why home store is not hourly availability. |
| **Customers** | Tickets, refunds and ratings, and how they relate to late deliveries. |
| **Root cause** | Six ranked hypotheses with evidence and charts, on all orders. |
| **Executive summary** | The one-minute story, findings, unresolved items and next steps. |
| **Key concepts and methodology** | Plain-language meanings of the terms used, each with a short example from this dataset. |
| **Analysis discussion** | Analysis overview and key questions with answers. |
| **Data audit** | Data dictionary, checks and decisions, and unresolved items. |

**Filters** (city, store, time of day, rain, date) apply to Delivery performance and Estimated stage times. Root cause and Executive summary always use all orders.

# 7. Findings, hypotheses and limitations

## Findings

1. The provisional North Star is 73.7% (1,092 ÷ 1,481), with a range of 72.8–73.7% depending on the 19 unclear cancellations. Among delivered orders, 75.1% are on time.
2. Severe delays are more common at peak. 25.0% of peak deliveries (7–10 PM) are more than 5 minutes late vs 11.8% off-peak. Late orders at peak run 9.19 min late on average vs 5.43 off-peak. The share late at all is not clearly different.
3. Promises don't match conditions. The average promise is 9.6 min at peak vs 10.5 off-peak, while actual time is 11.78 vs 11.20 min. In heavy rain the promise is no longer, and the late rate is higher (borderline).
4. Estimated rider wait is 1.02 min longer at peak while estimated picking is shorter. This is consistent with a handover bottleneck but does not prove one, and a rider shortage is not shown.
5. The stage estimates take almost the same share of time in late and on-time orders, so they cannot show which stage causes lateness.
6. Store load, store picking speed, new stores and customer tickets: no clear evidence in this analysis, which is not proof that they have no effect. Distance shows no clear difference, and traffic no clear straight-line association. Event days are less often late (14.3% vs 25.4%), which the data does not explain.

## Hypotheses, ranked

| Rank | Hypothesis | Evidence | Status |
|---|---|---|---|
| **1 H1** | Severe delays are more common at 7–10 PM. A store-to-rider bottleneck is one possible explanation. | Severely late 25.0% vs 11.8% (95% CI +7.8 to +18.6); holds in every distance and rain group and at 3 and 7 minutes. Late at all 29.0% vs 24.0%, not clear (95% CI −0.9 to +10.9). Estimated rider wait 1.02 min longer at peak; picking 0.53 min shorter. The bottleneck is a possible explanation, not an established cause; a rider shortage is not shown. | **Supported: severe delays are more common at peak** |
| **2 H2** | Promises do not match conditions: shorter at peak and no longer in heavy rain. | Peak promise 9.6 vs 10.5 min (95% CI −1.06 to −0.75); actual 11.78 vs 11.20 min. Heavy rain: promise 10.13 vs 10.36 min, late 33.9% vs 24.8% (95% CI 0.0 to +18.3). | **Supported at peak; borderline in rain** |
| **3 H3** | Store-hours busier than the store's usual load run late more often. | All hours 29.1% vs 21.4% (95% CI +1.4 to +14.0). Off-peak only 27.1% vs 21.7% (95% CI −1.5 to +12.3). Every peak hour has load ≥ 1.05. | **Inconclusive; overlaps with peak** |
| **4 H4** | Stores with slower picking deliver late more often. | r = 0.43 (95% CI −0.02 to +0.73, n = 20). Slower half 26.1% vs 23.9% (95% CI −2.3 to +6.6). Twenty stores is too few for a firm answer; this does not show picking speed has no effect. | **No clear evidence in this analysis** |
| **5 H5** | Late deliveries drive support tickets and refunds. | Tickets r = 0.02, refunds r = 0.08. No clear straight-line association. Tickets are per customer, not per order, so the test is weak and does not show lateness has no effect. | **No clear evidence in this analysis** |
| **6 H6** | New stores are less reliable. | 25.5% vs 24.9% (95% CI −6.8 to +8.0); only two new stores. No clear difference does not prove new and older stores perform the same. | **Inconclusive: only two new stores** |

95% CI = range of likely differences; if it includes zero, the difference is not clear. Also checked: distance (over 4 km 32.8% late vs up to 1 km 21.6%, no clear difference) and traffic (r = 0.02, no clear straight-line association).

## Limitations and unresolved items

- 19 cancellations (payment issue 8, customer cancelled 11) have no confirmed underlying cause, so the North Star is provisional. The recorded category does not establish the cause.

- We assume actual_delivery_at records confirmed customer handover and promised_delivery_at the original checkout promise, not an updated ETA. Both need to be confirmed.
- The Orders sheet holds 1.2% of the orders counted in Demand and under-represents peak hours (18.8% vs 27.9%). Whether these results apply more widely depends on how the sample was selected. That information is missing. Until it is known, results describe these 1,500 orders only, not the whole Blinkit network.
- Picking, rider-wait and travel durations are estimates, not recorded timestamps.
- There is no hourly rider availability; a rider's home store is not where they worked in a given hour.
- Support tickets and refunds are recorded per customer, not per order.
- The data covers two weeks (1–14 September 2026) and 20 stores, so store-level tests have little power.
- All findings are patterns, not proof of cause.

## Recommended next steps

1. Confirm the underlying cause of the unclear cancellations and finalise the North Star.
2. Get hourly rider log-ins and recorded pickup timestamps for 7–10 PM.
3. Review how the promised time is set at peak and in rain; pilot a buffer in a few stores.
4. Ask how the Orders sheet was sampled before drawing network-wide conclusions.

# 8. Key concepts and methodology

| Term | Meaning | Example from this dataset |
|---|---|---|
| **North Star** | The one number that best shows whether the goal is being met. Here it is the on-time fulfilment rate. | On-time orders ÷ eligible orders × 100 = 1,092 ÷ 1,481 × 100 = 73.7%. |
| **Eligible orders (E)** | The orders the North Star counts: every delivered order plus every company-caused cancellation. | 1,455 delivered + 18 rider unavailable + 8 item unavailable = 1,481. |
| **Provisional** | A result that may change once an open question is answered. | 19 cancellations have no confirmed cause yet. If all of them turned out to be company-caused, the North Star would be 72.8% instead of 73.7%. |
| **On time** | Handed over to the customer at or before the original checkout deadline. Assumption to confirm: actual_delivery_at records the confirmed handover, and promised_delivery_at is the original promise, never an updated ETA. | Promised 8:10 PM and handed over at 8:10 PM is on time. Handed over at 8:11 PM is late. |
| **Company-caused cancellation** | A cancellation whose recorded reason points to a failure on Blinkit's side. What counts is the underlying reason, not who pressed Cancel. | Rider unavailable (18) and item unavailable (8). "Customer cancelled" is only the recorded category: it does not tell us the cause, or even for certain who pressed Cancel, so those 11 orders stay unresolved. |
| **Severely late** | Delivered more than 5 minutes after the deadline. The 5-minute cut-off is our analytical choice, not a label in the data. | Promised 8:00 PM and delivered 8:06 PM is severely late. Delivered 8:04 PM is slightly late. |
| **Peak hours** | Orders placed from 7 PM to 10 PM (is_peak_hour in the data). All other hours are off-peak. | An order placed at 9:45 PM is peak; one placed at 10:05 PM is off-peak. |
| **P90** | The value that 90% of orders are at or below. It shows the slow tail that an average can hide. | P90 delivery time is 17.2 min: 9 in 10 delivered orders arrived within 17.2 minutes. |
| **Percentage points (pts)** | The simple difference between two percentages. | 25.0% at peak minus 11.8% off-peak = 13.2 percentage points. That is not the same as "13.2% higher". |
| **Correlation (r)** | A number from −1 to +1 showing how closely two measures rise or fall together in a straight line. A value near 0 means no clear straight-line association; it does not prove there is no relationship. | Store picking time vs store late rate: r = 0.43, but with only 20 stores the 95% CI runs from −0.02 to +0.73. |
| **95% confidence interval (CI)** | The range of differences that fit the data, given how many orders there are. If the range includes 0, we can't tell the two groups apart. | Severe-delay gap, peak minus off-peak: +13.2 pts, 95% CI +7.8 to +18.6. The whole range is above 0, so the gap is clear. |
| **Load ratio** | How busy a store-hour was compared with that store's usual hour: orders placed that hour (Demand sheet) ÷ typical orders per hour (Stores sheet). | A store that usually gets 20 orders an hour and gets 30 has a load ratio of 30 ÷ 20 = 1.5. |
| **Quartiles** | Four equal-sized groups made by sorting orders from lowest to highest. Q1 is the lowest quarter and Q4 the highest. | Off-peak delivered orders sorted by load ratio and split into four groups of about 295 orders each. |
| **Estimate** | A modelled value supplied in the data, not a measured timestamp. | picking_time_min_est is an estimate of how long picking took, not a clock reading. |
| **Sample** | Part of a larger group. Whether results from a sample apply to the larger group depends on how the sample was selected, not on its size alone. | The Orders sheet holds 1.2% of the orders in the Demand sheet. Whether these results apply more widely depends on how the sample was selected. That information is missing. |

| Term | Meaning | Example from this dataset |
|---|---|---|
| **Primary key** | The column that uniquely names each row. | order_id, store_id, rider_id, customer_id; Demand uses store_id + date + hour. |
| **Left join** | Keep every row of the first table and add matches where they exist. | All 1,500 orders stay; cancelled orders keep a blank rider. |
| **Reconciliation** | Checking that totals agree between sheets or after joining. | Stores sheet summaries equal the values recalculated from Orders. |

# 9. Key questions and answers

**1. Why is the North Star provisional?**

19 cancellations have no confirmed underlying cause and are left out of E. If all were company-caused, the North Star would be 72.8% instead of 73.7%.

**2. Why are rider unavailable and item unavailable counted as company-caused?**

Both recorded reasons describe a failure on Blinkit's side: no rider to deliver, or the item not in stock. The underlying reason decides this, not who pressed Cancel.

**3. Why is "Customer cancelled" not counted as company-caused?**

It is the recorded cancellation category. It does not establish the underlying cause, or even for certain who pressed Cancel; the customer may have cancelled because of a delay. The 11 orders stay unresolved until the reason is confirmed.

**4. What counts as "delivered on time"?**

Handover to the customer at or before the original checkout deadline. We assume actual_delivery_at records the confirmed handover and promised_delivery_at is the original promise, not an updated ETA. Both assumptions need to be confirmed.

**5. Why 5 minutes for "severe"?**

It is our analytical choice; the data has no severity label. The peak gap also holds at 3 and 7 minutes.

**6. Is the peak problem about lateness in general?**

Not clearly. 29.0% of peak deliveries are late vs 24.0% off-peak (95% CI −0.9 to +10.9). The difference is in how late they are.

**7. Does the data show a rider shortage at peak?**

No. Estimated rider wait is longer at peak, which is consistent with a bottleneck, but it is an estimate and there is no hourly rider availability.

**8. Why not use the stage times to find the slow step?**

They are estimates, and each stage takes almost the same share of time in late and on-time orders.

**9. Why was H3 (store load) not ranked higher?**

Every peak hour is a high-load hour. Within off-peak hours only, the gap is not clear (95% CI −1.5 to +12.3).

**10. Can these results describe all of Blinkit?**

Not on this evidence. Whether these results apply more widely depends on how the sample was selected. That information is missing. Peak hours are also under-represented compared with Demand, so for now results describe these 1,500 orders only.

**11. Does the Customers sheet disagree with Orders?**

No. orders_in_dataset counts delivered orders only; the gap equals the 45 cancellations.

**12. What data is needed next?**

Responsibility for the unclear cancellations, how the Orders sheet was sampled, hourly rider log-ins, and recorded pickup and handover timestamps.